5. The diameter of the bullock cart wheel is 1.4 *m*. Find the distance covered by it in 150 rotations?

6. A ground is in the form of a circle whose diameter is 350 *m*. An athlete makes 4 revolutions. Find the distance covered by the athlete.

7. A wire of length 1320 *cm* is made into circular frames of radius 7 *cm* each. How many frames can be made?

8. A Rose garden is in the form of circle of radius 63 *m*. The gardener wants to fence it at the rate of ₹150 per metre. Find the cost of fencing?

Objective type questions

9. Formula used to find the circumference of a circle is

(i) $2\pi r$ *units* (ii) $\pi r^2 + 2r$ *units* (iii) πr^2 *sq. units* (iv) πr^3 *cu. units*

10. In the formula, $C = 2\pi r$, '*r*' refers to

(i) circumference (ii) area (iii) rotation (iv) radius

11. If the circumference of a circle is 82π , then the value of 'r' is

(i) 41 cm (ii) 82 cm (iii) 21 cm (iv) 20 cm

12. Circumference of a circle is always

(i) three times of its diameter
(ii) more than three times of its diameter
(iii) less than three times of its diameter
(iv) three times of its radius

2.4 Area of the Circle

Let us consider the following situation.

A bull is tied with a rope to a pole. The bull goes round to eat grass. What will be the portion of grass that the bull can graze?

Can you tell what is needed to be found in the above situation, Area or Perimeter? In this situation we need to find the area of the circular region.

Let us find a way to calculate the area (A) of a circle in terms of known area, that is the area of a rectangle.

1. Draw a circle on a sheet of paper.

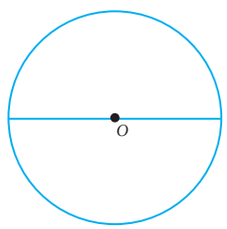


Fig. 2.12

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